

# Introduction to Contract Theory and Auctions

## The Classical Auction Formats under Complete Information

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- In an auction, a “good” is sold to the party who submits the highest bid.
- Broadly defined, auctions are used to allocate economic resources: works of art, government bonds, natural resource rights (oil, minerals, fishing, etc.), radio spectrum, and many others.
- They have many formats/rules:
  - Bids can be submitted sequentially or in sealed envelopes.
  - The price paid may be the highest bid or some other price.
  - If selling different units, bids maybe taken simultaneously on all units, or each unit maybe sold sequentially.
- Auctions can be modeled as strategic games.
  - Usually as games of incomplete information due to the lack of complete information about important features of the environment.
- A game-theoretic analysis helps understand the consequences of various auction designs/rules, assess the format that is most effective at allocating resources, and the one likely to raise the most revenue.

- $n$  players.
- Each player with fixed valuation  $v_i > 0$  for some item being auctioned.
- Assume, without loss of generality, that each players' valuation for the object are all different; further we number the players from 1 to  $n$  in such a way that  $v_1 > v_2 > \dots > v_n > 0$ .
- Assume that this valuation configuration is **common knowledge** → there is complete information about the players' valuations.
- A player  $i$  submits a bid  $b_i$ .
- The rules of the auction determine, given a bid profile  $b = (b_1, \dots, b_n)$ , how the winner is determined, and how much each player needs to pay.
- In the classical auction formats (first-price, second-price, English auction), the winner is the player who submitted the highest bid.

## Sealed-Bid Second Price Auction

- In this auction, a player submits a bid  $b_i$ .
- The player who submitted the highest bid  $b_{\max}(b) = \max_j b_j$  wins.
- However, the winner pays the second highest bid  $b_{\text{nd}}(b)$ . If player  $i$  wins, then  $b_{\text{nd}}(b) = \max_{j \neq i} b_j$ .
- Payoffs are then:

$$u_i(b) = \begin{cases} v_i - b_{\text{nd}}(b) & \text{if } b_i = b_{\max}(b) \\ 0 & \text{otherwise} \end{cases}$$

- In case player  $i$  is in a tie for the highest bid, her payoff depends on how the ties are broken.
  - Assume that in this case, the winner is the player among those submitting the highest whose “number” is smallest (i.e., that player with the highest valuation).

## Equilibria of the Sealed-Bid Second Price Auction

- The game has many Nash equilibria.
- $(b_1, \dots, b_n) = (v_1, \dots, v_n)$ , i.e., each player's bid is her valuation for the object.
  - Since  $v_1 > v_2 > \dots > v_n$ , the outcome is that player 1 obtains the object at price  $b_2 = v_2$ , and her payoff is  $v_1 - v_2$ , every other player's payoff is zero.
  - Why it is an equilibrium? If player 1 changes her bid to some other price at least equal to  $b_2$ , then the outcome does not change. If she changes her bid to a price less than  $b_2$ , then she loses and obtains zero.
  - If some other player lowers her bid or raises it to some price at most equal to  $b_1$ , then she remains a loser, if she raises her bid above  $b_1$ , then she wins, but in paying  $b_1$ , makes a loss.
- $(b_1, \dots, b_n) = (v_1, 0, \dots, 0)$  is another equilibrium.
  - If player 1 changes her bid, then the outcome remains the same.
  - If any of the remaining players raises her bid, then either the outcome remains the same (if her new bid is at most  $v_1$ ) or causes her to obtain the object at a price that exceeds her valuation (if her bid exceeds  $v_1$ ).
  - Player 1 obtains the object at a price of 0!

## Equilibria in the Second Price Auction

- There are other equilibria in which player 1 does not obtain the object.
- Consider  $b = (v_2, v_1, 0, \dots, 0)$  in which player 2 obtains the object at price  $v_2$  and every player (including player 2) obtains a payoff of zero.

- In the previous equilibrium, player 2's equilibrium bid exceeds her valuation: if player 1 were to increase her bid to any value less than  $v_1$ , player 2's payoff would be negative.
- This remark doesn't affect the status of the profile  $b = (v_2, v_1, 0, \dots, 0)$  as an equilibrium, because in a Nash equilibrium, players only consider **unilateral deviations**, i.e., they do not consider the risk that other player will take an action different from their equilibrium action, each player chooses an optimal action given the equilibrium profile.
- However, the previous remark shows that there is something “off” with the previous equilibrium which makes it less plausible.

## Equilibria in the Second Price Auction

- The weakness of the equilibrium  $b = (v_2, v_1, 0, \dots, 0)$  is because player 2's bid  $v_1$  is **weakly dominated** by the bid  $v_2$ .

### Proposition 2.1

*In a second-price sealed-bid auction (with complete information), a player's bid equal to her valuation weakly dominates all her other bids.*

- For any bid  $b_i \neq v_i$ , player  $i$ 's bid  $v_i$  is at least as good as  $b_i$ , **no matter what the other players bid**.
- The key point: in a second-price sealed bid auction, a player cannot change the price she pays, but only possibly changes her status from a winner to a loser, or vice versa.
- Summarizing, the second-price auction has many Nash Equilibria, but the equilibrium  $(b_1, \dots, b_n) = (v_1, \dots, v_n)$  is distinguished by being the only one in weakly dominant strategies.



## Sealed-Bid First Price Auction

- In this auction, a player submits a bid  $b_i$
- The player who submitted the highest bid  $b_{\max}(b) = \max_j b_j$  wins.
- The winner pays her own bid  $b_{\max}(b)$ .
- Payoffs are then:

$$u_i(b) = \begin{cases} v_i - b_{\max}(b) & \text{if } b_i = b_{\max}(b) \\ 0 & \text{otherwise} \end{cases}$$

- In case player  $i$  is in a tie for the highest bid, her payoff depends on how the ties are broken.
  - Assume that in this case, the winner is the player among those submitting the highest whose “number” is smallest (i.e., that player with the highest valuation).

## Equilibria in the First-Price Auction

- The game has many Nash Equilibria
- $(b_1, \dots, b_n) = (v_2, v_2, v_3, \dots, v_n)$  that is, player 1's bid is player 2's valuation  $v_2$ , and every other player's bid is her own valuation. The outcome of this equilibrium is player 1 obtains the object at the price  $v_2$ .

- There are many other equilibria, but in all of them, the winner is the player who values the object most highly, i.e., player 1.
  - In any action profile  $(b_1, \dots, b_n)$  in which some player  $i \neq 1$  wins, we have  $b_i > b_1$ . If  $b_i > v_2$ , then  $i$ 's payoff is negative, so that she can do better by reducing her bid to 0.
  - If  $b_i \leq v_2$ , then player 1 can increase her payoff from 0 to  $v_1 - b_i$  by bidding  $b_i$  in which case she wins. Thus no such profile can constitute an equilibrium.
- Actually, in a Nash Equilibrium of the first-price sealed-bid auction, the two highest bids are the same, one of these bids is submitted by player 1, and the highest bid is at least  $v_2$  and at most  $v_1$ .

# Equilibria in the First-Price Auction

- In any equilibrium in which the winning bid is larger than  $v_2$ , at least one player's bid exceeds her valuation.
- Although this seems risky if such a player were to win, at an equilibrium, there is no risk → remember unilateral deviations!
- Since equilibrium the bid doesn't win, there is no "risk". However, this issue reduces the "plausibility" of this type of equilibria.
- Similarly to the second price auction, this "riskiness" to player  $i$  of a bid  $b_i > v_i$  is reflected in the fact that it is **weakly dominated** by bid  $v_i$ .
  - If the other players' bids are such that player  $i$  loses, when she bids  $b_i$ , then the outcome is the same, whether she bids  $b_i$  or  $v_i$ .
  - If the other players' bids are such that player  $i$  wins when she bids  $b_i$ , then her payoff is negative when she bids  $b_i$ , and zero when she bids  $v_i$  (regardless of whether this bid wins).

- However, a bid  $b_i < v_i$  of player  $i$  is not weakly dominated by the bid  $v_i$ .
- Actually, such a bid is not weakly dominated by any bid.
- Also, even if  $v_i$  weakly dominates higher bids, this bid is itself dominated by a lower bid.

## Proposition 2.2

*In a first-price sealed-bid auction (with complete information), a player's bid of at least her valuation is weakly dominated, and a bid of less than her valuation is not weakly dominated.*

- In a first price auction, at least one player's action is weakly dominated. However, this depends on the assumption that a bid may be any number.

## Restricting the bids in a First-Price Sealed Bid Auction

- Consider the variant of the FPSA in which bids and valuations are restricted to be multiples of some discrete monetary unit:  $\epsilon$ .

- Then an action profile

$$(v_2 - \epsilon, v_2 - \epsilon, b_3, \dots, b_n)$$

where  $b_j \leq v_j - \epsilon$  for  $j = 3, \dots, n$ , is a Nash equilibrium in which **no player's bid is weakly dominated**.

- Furthermore, every equilibrium in which no player's bid is weakly dominated takes this form.
- If  $\epsilon \rightarrow 0$ , each equilibrium above is “close” to an the equilibrium  $(v_2, v_2, b_3, \dots, b_n)$ ,  $b_j \leq v_j$ , for the game with unrestricted bids.
- For this reason, we call the equilibria  $(v_2, v_2, b_3, \dots, b_n)$  with  $b_j \leq v_j$  the **distinguished equilibrium of the FPSA**.

- Although both auction formats have multiple Nash Equilibria, their distinguished equilibria yield the same outcome: in both auction formats, the good is sold to player 1 and the revenue to the auctioneer is  $v_2$ .
- When we restrict attention to this distinguished equilibria, both formats are **revenue equivalent**.
  - Such a result actually holds more generally: under some conditions, **all auction formats will yield the same expected revenue**.



## All-pay auctions

- Suppose there are only two bidders, and that each of them bids for a perfectly divisible commodity that is worth  $\$V$  to each of them.
- Once again, the winner is bidder whose bid is higher.
- In the case of a tie, each person receives half of the commodity, which she values at  $\$V/2$ .
- In contrast with the previous auction formats, each person now needs to pay her own bid, **regardless of whether she wins**.
- This situation can be modeled as a game between two players, where:
  - Each player's set of actions is the set of possible bids:  $[0, V]$ .
  - Payoff for player  $i$  are given by:

$$u_i(b_1, b_2) = \begin{cases} V - b_i & \text{if } b_i > b_j \\ V/2 - b_i & \text{if } b_i = b_j \\ -b_i & \text{if } b_i < b_j \end{cases}$$

where  $j$  is the other player.

- What situations can be modeled as an all-pay auction?
  - Lobbying
  - R&D races.

- An all-pay auction has no pure strategy Nash equilibrium.
  - ① No pair of bids  $(b, b)$  with  $b < V$  is a Nash equilibrium: either player can increase her payoff by slightly increasing her bid.
  - ②  $(V, V)$  is not a NE because either player can increase her payoff from  $-V/2$  to 0 by reducing her bid to 0.
  - ③ No pair of bids  $(b_1, b_2)$  with  $b_1 \neq b_2$  is a NE because the player whose bid is higher can increase her payoff by reducing her bid. On the other hand, the player whose bid is lower can increase her payoff by reducing her bid to 0.
- Therefore we will need to look for mixed strategy Nash equilibria.
  - Given that the action space is an interval, how would we define a mixed strategy?
  - How to characterize a mixed strategy equilibrium when there is a continuum of actions?

## Mixed strategies in games with a continuum of actions.

- Recall that for games with **finitely many actions**, a profile  $\alpha^*$  is a Mixed Strategy Nash Equilibrium if and only if for each player  $i$ 
  - the expected payoff, given  $\alpha_{-i}^*$ , to every action to which  $\alpha_i^*$  assigns positive probability is the same
  - the expected payoff, given  $\alpha_{-i}^*$ , to every action to which  $\alpha_i^*$  assigns zero probability is at most the expected payoff to any action to which  $\alpha_i^*$  assigns positive probability.
- Unfortunately, for games with a **continuum of actions**, the above definition does not work because of (1)
  - A mixed strategy in such a game is determined by the probabilities it assigns to sets of actions, and not the probability it assigns to single actions (which may be zero).
  - Intuitively, recall that for a random variable  $X$  with a **continuous** density  $f$ , the probability that  $X = x$ , where  $x$  is some value, is zero.

## Proposition 2.3

*A mixed strategy profile  $\alpha^*$  in a strategic game is a mixed strategy NE if and only if, for each player  $i$ ,*

- ①  $\alpha_i^*$  assigns probability zero to the set of actions  $a_i$  for which the action profile  $(a_i, \alpha_{-i}^*)$  yields player  $i$  an expected payoff less than her expected payoff to  $\alpha_i^*$*
- ② for no action  $a_i$ , does the action profile  $(a_i, \alpha_{-i}^*)$  yield player  $i$  an expected payoff greater than her expected payoff to  $\alpha_i^*$ .*

*In math, for every player  $i$ :*

$$\text{supp}(\alpha_i^*) \subseteq \arg \max_{a_i \in A_i} \mathbb{E}[u_i(a_i, \alpha_{-i}^*)]$$

*where  $\text{supp}(\alpha_i^*)$  denotes the support of  $\alpha_i^*$ , i.e., the set of actions of player  $i$  with positive probability according to  $\alpha_i^*$ .*

## Mixed strategies in games with a continuum of actions

- A mixed strategy in such a game for player  $i$  can be represented by a **cumulative distribution function**  $F_i$ .
- In games in which player's actions are one-dimensional intervals of numbers, say  $\underline{a}_i$  and  $\bar{a}_i$ , a mixed strategy for player  $i$  is a non-decreasing function  $F_i$  for which  $0 \leq F_i(a_i) \leq 1$  for every action  $a_i \in [\underline{a}_i, \bar{a}_i]$ .
- The form of a mixed strategy NE in a game with a continuum of actions can have a very complex shape.
- There are some games in which the mixed strategy equilibria have a very simple shape:
  - The mixed strategy assigns probability zero to all actions except those in an interval.
  - There are numbers  $x_i$  and  $y_i$  such that  $i$ 's mixed strategy assigns:  $F_i(z) = 0$  if  $z < x_i$  and  $F_i(z) = 1$  for  $z \geq y_i$ .
  - Player  $i$ 's expected payoff when her action is  $a_i$  and the other players use mixed strategies  $F_j$ ,  $j \neq i$ , takes the form:

$$\begin{cases} = c_i & x_i \leq a_i \leq y_i \\ \leq c_i & a_i < x_i \text{ or } a_i > y_i \end{cases}$$

## Back to the all-pay auction

- We will look for an equilibrium in which no mixed strategy assigns positive probability to **any single bid**.
- $F_i(b_i)$  is the probability that player  $i$  bids at most  $b_i$  and also the probability that she bids less than  $b_i$ .
- We will also look for strategy pairs  $(F_1, F_2)$  for which there are numbers  $x_i$  and  $y_i$ ,  $i = 1, 2$  such that  $F_i$  assigns positive probability only on the interval  $[x_i, y_i]$ .
- Consider player 1's expected payoff when she uses bid  $b_1$ , given player 2's mixed strategy  $F_2$ :
  - If  $b_1 < x_2$ , then  $b_1$  is less than player 2's bid with probability 1, so P1's payoff is  $-b_1$ .
  - If  $b_1 > y_2$ , then  $b_1$  exceeds player 2's bid with probability 1, so P1's payoff is  $V - b_1$ .
  - If  $x_2 < b_1 < y_2$ , then P1's expected payoff is:
    - With prob.  $F_2(b_1)$  P2's bid is less than  $b_1 \rightarrow$  P1's payoff is  $V - b_1$ .
    - With prob.  $1 - F_2(b_1)$  P2's bid exceeds  $b_1, \rightarrow$  P1's payoff is  $-b_1$ .
    - The probability that P2's bid is exactly  $b_1$  is zero.
    - Thus P1's expected payoff is  $(V - b_1)F_2(b_1) + (-b_1)(1 - F_2(b_1)) = VF_2(b_1) - b_1$ .

## Mixed strategy equilibrium in the all-pay auction

- We need to find values  $x_2$  and  $y_2$  and a strategy  $F_2$ , such that player 1's expected payoff is:
  - constant in the interval  $x_1$  to  $y_1$ ;
  - less than this constant when  $b_1 < x_1$  and  $b_1 > y_1$ .
- Thus, we need:
  - $VF_2(z) - z = c_1$  for  $x_1 \leq z \leq y_1$  for some constant  $c_1$ .
  - $F_2(x_2) = 0$  and  $F_2(y_2) = 1$ .
- Analogous conditions must be satisfied for  $F_1$  :
  - $VF_1(z) - z = c_2$  for  $x_2 \leq z \leq y_2$  for some constant  $c_2$ .
  - $F_1(x_1) = 0$  and  $F_1(y_1) = 1$ .